

Employment

Graduate Research Assistant	NC State University	Apr 2025 – Present
------------------------------------	----------------------------	---------------------------

- Designed and deployed a cloud-native ML data platform on GCP (Cloud Run) integrating JBrowse 2 genome browser, enabling feature extraction and model training on 500GB+ genomic datasets for 100+ researchers.
- Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.

Product Engineer Intern	Isomer AI	May 2025 – May 2026
--------------------------------	------------------	----------------------------

- Shipped per-tenant AI workflow platform for insurance carriers: Temporal.io durable orchestration, LLM risk signal detection across Claude/GPT-4o/Gemini, Auth0 RBAC, MCP servers; 30+ deployments.
- Built MCP servers (Python + TypeScript); Playwright + Tailscale VPN for customer-internal browser automation.

Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
----------------------------------	-----------------------------	----------------------------

- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML models, improving document processing accuracy by 25% across high-volume enterprise insurance pipelines.
- Automated AWS and GCP CI/CD pipelines; integrated Grafana monitoring and led model validation and red-teaming sessions.

AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
---------------------	------------------	----------------------------

- Led full SDLC for a data-driven support ticket analysis system using NLP classification models, reducing average resolution time by 15% and improving first-contact resolution rates across enterprise clients.
- Built Kafka + Spark + Hadoop pipelines with data engineers; cut ticket-processing latency by 20% at peak load.

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
--------------------	--	----------------------------

- Master of Science in Computer Science
- Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.

Mumbai, India	Mumbai University	Aug 2020 – May 2024
----------------------	--------------------------	----------------------------

- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.

Projects

Inventory Reservation Service	Java 21 + Spring Boot REST API	2026
--------------------------------------	---------------------------------------	-------------

- Built a Java 21 + Spring Boot inventory reservation API: PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations, idempotent POST endpoints, automated expiry via scheduled cleanup job.
- Testcontainers integration tests verified concurrency: qty-1 race produces exactly one success, one 409 Conflict.

KubeServe	Kubernetes LLM Inference Platform	2025
------------------	--	-------------

- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.
- Deployed via Helm + ArgoCD GitOps on Kubernetes; load tested k6 to 500 VUs; 88 pytest integration tests.

PulseRank	Real-Time Leaderboard & Activity Feed	2026
------------------	--	-------------

- Built event-driven leaderboard and activity feed in TypeScript with NestJS and Next.js: MySQL transactional writes, MongoDB feed projections with TTL, Redis sorted sets for sub-millisecond reads; live SSE.
- Validated multi-DB consistency with Testcontainers; benchmarked hot paths with k6 at 10K concurrent users.

Languages and Technologies

- Languages:** Java, Python, TypeScript, JavaScript, SQL, Ruby
- AI/ML:** LLMs (Claude, GPT-4o, Gemini), MCP tooling, Temporal.io, RAG, embeddings, PyTorch, scikit-learn
- Frameworks:** Spring Boot, FastAPI, Node.js, NestJS, Next.js, React, React Native, Express
- Cloud & Infra:** AWS, GCP, PostgreSQL, Redis, Kafka, Docker, Kubernetes, Terraform, GitHub Actions

Additional Experience and Awards

- Open Source:** Contributed to Expertiza Rails platform (5,000+ users, 300+ contributors); added 110+ RSpec tests, increased coverage by 25%, reduced redundancy by 35%.
- Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; shipped 3 full-stack MVPs end-to-end within 48-hour competitive sprints.
- Publications:** Co-authored CV-Based Cybersecurity Threat Detection (Springer 2023) and Mapify (IEEE 2022); presented to 500+ global attendees.
- GDSC Lead:** Led Google Developer Student Club; organized 4+ tech events (hackathons, conferences) reaching 500+ students.